

Valgrind:

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riets@LAPTOP-RAGP3AP3:/mnt/c/Users/botha/Downloads/COS-214-Practical-4-main/COS-214-Practical-4-main$ docker run -it --rm -v "${pwd}":/workspace -w /workspace taskforge-env bash
root@f41a82d426b:/workspace# make valgrind
g++ -std=c++11 -g *.cpp -o taskforge
valgrind --leak-check=full --show-leak-kinds=all ./taskforge
==24== Memcheck, a memory error detector
==24== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==24== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==24== Command: ./taskforge
==24==

=== Scenario 1: coordinated power-plant fire response ===
The city manager dispatches a nested incident hierarchy.
New emergency: Tshwane emergency command
New emergency: Central district response
New emergency: Industrial sector response
New emergency: Power-plant fire incident
    INDIVIDUAL VEHICLE SENT OUT FOR Fire engine 1 to the turbine hall
    PRIORITY P1 - trapped workers: dispatch first for Rescue ambulance to the turbine hall
    INDIVIDUAL VEHICLE SENT OUT FOR Rescue ambulance to the turbine hall
Completed Power-plant fire incident
Completed Industrial sector response
Completed Central district response
Completed Tshwane emergency command

Depth-first incident view:
- Central district response
- Industrial sector response
- Power-plant fire incident
- Fire engine 1 to the turbine hall
- Rescue ambulance to the turbine hall [priority: P1 - trapped workers] [audited]

Reverse incident view:
- Central district response

Audit trail for the stacked dispatch decorators:
  1. Dispatch started: Rescue ambulance to the turbine hall [priority: P1 - trapped workers]
  2. Dispatch completed: Rescue ambulance to the turbine hall [priority: P1 - trapped workers]

=== Scenario 2: new hazard during an active incident ===
A depth-first snapshot is created before a hazmat team is added.
Existing snapshot sees the new hazmat unit: no
Fresh snapshot sees the new hazmat unit: yes

Updated depth-first incident view:
- Fire engine 1 to the turbine hall
- Rescue ambulance to the turbine hall [priority: P1 - trapped workers] [audited]
- Hazmat unit to chemical storage [priority: P1 - escalating chemical risk] [audited]

Dispatching the updated power-plant incident:
New emergency: Power-plant fire incident
    INDIVIDUAL VEHICLE SENT OUT FOR Fire engine 1 to the turbine hall
    PRIORITY P1 - trapped workers: dispatch first for Rescue ambulance to the turbine hall
    INDIVIDUAL VEHICLE SENT OUT FOR Rescue ambulance to the turbine hall
    PRIORITY P1 - escalating chemical risk: dispatch first for Hazmat unit to chemical storage
    INDIVIDUAL VEHICLE SENT OUT FOR Hazmat unit to chemical storage
Completed Power-plant fire incident
[STATE] Emergency evaluation unit dispatched: Pending -> Dispatched
[STATE] Emergency evaluation unit en route: Dispatched -> En Route
[STATE] Emergency evaluation unit on scene: En Route -> On Scene
[REJECTED] Cannot cancel (response already committed on scene) "Emergency evaluation unit" while in state: On Scene
[STATE] Emergency evaluation unit completed: On Scene -> Completed
[REJECTED] Cannot arrive on scene (not dispatched yet) "State rejection" while in state: Pending
[REJECTED] Cannot complete (not dispatched yet) "State rejection" while in state: Pending
[STATE] State rejection cancelled: Pending -> Cancelled
[REJECTED] Cannot dispatch (task was cancelled) "State rejection" while in state: Cancelled
[REJECTED] Cannot arrive on scene (task was cancelled) "State rejection" while in state: Cancelled
[REJECTED] Cannot complete (task was cancelled) "State rejection" while in state: Cancelled
[REJECTED] Cannot cancel (already cancelled) "State rejection" while in state: Cancelled
[STATE] State progression dispatched: Pending -> Dispatched
[REJECTED] Cannot dispatch (already dispatched) "State progression" while in state: Dispatched
[REJECTED] Cannot complete (not on scene yet) "State progression" while in state: Dispatched
[STATE] State progression en route: Dispatched -> En Route
[REJECTED] Cannot dispatch (already dispatched) "State progression" while in state: En Route
[REJECTED] Cannot complete (not on scene yet) "State progression" while in state: En Route
[STATE] State progression on scene: En Route -> On Scene
[REJECTED] Cannot arrive on scene (already on scene) "State progression" while in state: On Scene
[STATE] State progression completed: On Scene -> Completed
[REJECTED] Cannot dispatch (task already completed) "State progression" while in state: Completed
[REJECTED] Cannot arrive on scene (task already completed) "State progression" while in state: Completed
[REJECTED] Cannot complete (already completed) "State progression" while in state: Completed
[REJECTED] Cannot cancel (task already completed) "State progression" while in state: Completed

Taskforge emergency response scenarios completed successfully.
==24==
==24== HEAP SUMMARY:
==24==    in use at exit: 0 bytes in 0 blocks
==24==   total heap usage: 196 allocs, 196 frees, 83,034 bytes allocated
==24==
==24== All heap blocks were freed -- no leaks are possible
==24==
==24== For lists of detected and suppressed errors, rerun with: -s
==24== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
root@f41a82d426b:/workspace#
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